

U2210251

Javohirbek Xatamov

Section 004

Waveform 1

Period: $T = 2 \text{ ms} = 0,002 \text{ s}$

Fundamental frequency: $\omega_0 = \frac{2\pi}{T} =$

$$= 1000\pi \text{ rad/s}$$

Symmetry: Even function as

$$(f(t) = f(-t)) \text{ so } b_n = 0$$

① Average value (a_0):

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt = \frac{1}{0,002} \int_{-0,0005}^{0,0005} 1 dt = \frac{0,001}{0,002} =$$

$$= \frac{1}{2}$$

② Cosine coeff's (a_n):

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t) dt =$$

$$= \frac{2}{0,002} \int_{-0,0005}^{0,0005} \cos(1000\pi n t) dt =$$

$$= \frac{2}{0,002} \left[\frac{\sin(1000\pi n t)}{1000\pi n} \right]_{-0,0005}^{0,0005} =$$

$$\Rightarrow \frac{\sin(1000\pi n \cdot 0,0005) - \sin(1000\pi n \cdot (-0,0005))}{1000\pi n} =$$

$$= \frac{2}{0,002} \cdot \frac{2 \sin(0,5 n \pi)}{1000\pi n} = \frac{2}{n\pi} \cdot \sin\left(\frac{n\pi}{2}\right)$$

$$a_n = \frac{2}{n\pi} \cdot \sin\left(\frac{n\pi}{2}\right)$$

Waveform 2

~~Period~~ $T = 0,004 \text{ s}$

High pulse: $t \in [-0,5 \text{ ms}, 0,5 \text{ ms}]$

Even function as $f(x) = f(-x)$ so

$$b_n = 0$$

① Average Value (a_0):

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt = \frac{1}{0,004} \int_{-0,0005}^{0,0005} 1 dt = \frac{0,0005}{0,004} =$$

$$= \frac{-0,0005}{0,004} = \frac{0,001}{0,004} = \frac{1}{4}$$

② Cosine Coeff's (a_n):

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t) dt = \frac{2}{0,004} \int_{-0,005}^{0,005}$$

$$\cos(500\pi n t) dt = \left[\frac{\sin(500\pi n t)}{500\pi n} \right]_{-0,005}^{0,005} =$$

$$= \frac{\sin(500\pi n \cdot 0,0005) - \sin(500\pi n(-0,0005))}{500\pi n}$$

$$\Rightarrow [\text{as } \sin(-x) = -\sin(x)] \Rightarrow$$

$$\Rightarrow \frac{2 \sin(0,25\pi n)}{500\pi n} \Rightarrow \frac{2}{0,004} \cdot \frac{2 \sin(0,25\pi n)}{500\pi n}$$

$$= \frac{2}{\pi} \sin\left(\frac{n\pi}{4}\right)$$